

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

DATE: \_\_\_\_\_

## Fertilizer in Two Greenhouses

AP Statistics · Unit 1 · Topic 1.13 · B: 2026-10-05 · E: 2026-10-16

### [STAR] TODAY'S PROBLEM

Suppose a grower has 24 similar tomato plants, 12 in a sunny greenhouse and 12 in a shaded greenhouse. The response is fruit yield in kilograms per plant.

**Design A:** Completely randomize all 24 plants; assign 12 to fertilizer and 12 to no fertilizer, ignoring greenhouse.

**Design B:** Treat greenhouse as a block; within each greenhouse, randomly assign 6 plants to fertilizer and 6 to no fertilizer.

Pilot means from plants grown under the same conditions are shown.

#### Mean fruit yield (kg per plant)

| Greenhouse | Fertilizer | None |
|------------|------------|------|
| Sunny      | 7.2        | 6.0  |
| Shaded     | 4.0        | 2.8  |

### FIRST TAKE — before the video (5 min)

Which design would you use to detect the fertilizer effect most clearly? Name one detail behind your choice.

### [BOOK] BLOCK, RANDOMIZE, THEN INFER (keep on the page)

In a **completely randomized design**, treatments are assigned across all units. In a **randomized block design**, first group units that are similar on a relevant variable, then randomly assign treatments *within each block*. Blocking separates natural block-to-block variability from the treatment comparison. Random assignment supports cause only when the observed difference is too large for chance alone; generalization also requires units that represent the target population.

**Finish after the video:**

**DOK 1** (a) Name the blocking variable and the response variable, with units.

**DOK 2** (b) For the sunny greenhouse, find fertilizer minus no fertilizer. For the fertilized plants, find sunny minus shaded. Compare the two differences.

**DOK 3** (c) Which design should the grower use? Use the two differences to explain your choice and how chance imbalance could affect Design A. If Design B gives a statistically significant difference, state a causal conclusion limited to the plants and conditions studied.

**Frame:** Design \_\_\_\_\_ is stronger because the block difference is \_\_\_\_\_ kg versus a treatment difference of \_\_\_\_\_ kg.

**Frame:** If significant, the result supports \_\_\_\_\_ for \_\_\_\_\_, but not \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_